The World Commonwealth Index (WCI): Open, Auditable Wellbeing Framework and Reference-Unit Research Programme

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1 Abstract

This paper specifies the World Commonwealth Index (WCI): an open, auditable, non-income composite index of societal wellbeing built on five weighted pillars — Family and Human Development (50%), Health (15%), Civil Peace and Security (15%), Ecology (10%), and Trust and

Institutional Integrity (10%). Country scores are produced by decennial goalpost normalization with clipping, coverage-gated within-pillar aggregation, a weighted geometric mean with a unit floor that limits compensability across pillars, and three quadratic severe-harm penalties covering trafficking and bonded child labour, state-attributable civilian casualties, and biospheric toxicity, with a mechanical deduction ceiling of 27.06%. Two companion diagnostics — the Sustainable Improvement Factor and the Wellbeing Over Income index — are specified alongside.

2 Introduction and Motivation

Since Kuznets first cautioned that national income should not be conflated with national welfare (Kuznets, 1934), a substantial literature has documented the gap between material output and human flourishing (Easterlin, 1974). The Commission on the Measurement of Economic Performance and Social Progress crystallized this critique into a research agenda: shift measurement from production to wellbeing, and from aggregates to the joint distribution of outcomes (Stiglitz, Sen and Fitoussi, 2009). Capability theory supplied the conceptual foundation (Sen, 1985; 1999; Anand and Sen, 1994), and composite instruments such as the Human Development Index and the OECD–EC JRC handbook on composite indicators operationalized parts of it (UNDP, Human Development Report; Nardo et al., 2008). Yet no existing index places the durability of the family, freedom from severe harms, and institutional trust at the normative centre of a non-compensatory, capture-resistant measurement architecture — the gap the present specification is designed to address (WCI Master Specification, §1; WCI Unified Specification, §1).

The World Commonwealth Index (WCI) is proposed to fill that gap. It is deliberately and explicitly **not a currency, not a peg, not a claim on any asset, and not a proposal to amend the IMF's Special Drawing Rights**; the term "index" is used in the statistical sense. It is a reference unit in the research sense: a common, transparent yardstick against which the true wealth of societies — health, peace, family integrity, ecological safety, and trust — can be measured and compared, independently of the volatility and asymmetries of dominant national currencies that the literature on reserve-currency dilemmas has long highlighted (Triffin, 1960).

This document is released as an open specification under CC BY 4.0 (text) and Apache 2.0 (code), and as a standing invitation to global scholarly collaboration.

3 The Five-Pillar Architecture

The WCI aggregates five pillars, each composed of internationally comparable indicators with published sources (UN IGME, 2024; UNODC, 2023; WHO, 2021; 2024; V-Dem, 2024; World Bank, World Development Indicators; EDGAR, 2024; Gallup World Poll).

Pillar 1 — Family and Human Development (weight 0.50). Child survival, child poverty, education completion, and protection of children and households from violence (UN IGME, 2024; World Bank, WDI). The normative claim behind the heaviest weight is sociological: chronic psychological distress, addiction, crime, and violence cluster around the dissolution of stable, supportive family environments; a genuine wellbeing index should therefore measure family integrity not as one cultural preference among many, but as the primary substrate of human development.

Pillar 2 — Health (weight 0.15). Healthy life expectancy (HALE), maternal mortality, and uni-

versal health coverage (UHC) service indices (WHO, 2024).

Pillar 3 — Civil Peace and Security (weight 0.15). Homicide rates, exposure to armed conflict, and arbitrary detention (UNODC, 2023).

Pillar 4 — Ecology and Basic Resources (weight 0.10). Fine particulate exposure (PM2.5), ecosystem integrity, and land-cover conversion (EDGAR, 2024).

Pillar 5 — Trust, Freedom and Institutional Integrity (weight 0.10). Institutional trust, control of corruption, civic freedom, and access to reliable information (V-Dem, 2024).

The weights (0.50, 0.15, 0.15, 0.10, 0.10) are a normative commitment, published openly and subject to ratification by the Citizens' Assembly described below — they are not presented as statistically estimated parameters, and the framework provides the sensitivity machinery (Section on robustness) for others to test alternative weightings.

4 Measurement Methodology and Mathematical Specification

This section consolidates the mathematical core of the framework.

4.1 Decennial goalpost normalization

For a positive indicator with lower and upper goalposts L and U fixed for each decennial release cycle, the normalized value is $q = 100 \cdot \text{clip}\left(\frac{x-L}{U-L},\ 0,\ 1\right)$; for negative indicators the transform is reversed. Saturation fractions are reported, and goalposts are revised only at decennial reviews with published overlap series, so that cross-year comparability is preserved.

4.2 Within-pillar aggregation and eligibility

Within pillar i, let $\mathcal{A}_i$ be the set of indicators with adequate data for country c at time t. The pillar score is

$$P_i = \text{clip}\left[\left(\sum_{j\in\mathcal{A}_i} a_{ij}\, q_{ijt}\, r_j + \sum_{j\in\mathcal{A}_i} a_{ij}\, \tilde{q}_{ijt}\,(1-r_j)\right)\cdot m_i,\ 1,\ 100\right],$$

where a_{ij} are within-pillar weights, r_j indicates survey versus administrative sourcing, $\tilde{q}$ denotes lower-quartile imputation for sensitive items under a single preregistered imputation convention, and m_i is the observed weighted coverage multiplier. A country is eligible only with at least 70% overall coverage and at least 50% coverage in every pillar.

4.3 Cross-pillar aggregation: non-compensatory geometric mean

The unpenalized index is

$$W = 100 \times \prod_{i=1}^{5}\left(\frac{\max(P_i, 1)}{100}\right)^{w_i}, \qquad \sum_i w_i = 1.$$

The unit floor prevents undefined logarithms when a pillar collapses while still imposing a heavy multiplicative penalty on imbalance; the compression of the bottom tail this introduces is disclosed as a known property.

4.4 Severe-harm penalties

Three harms are treated as non-compensable in degree through quadratic penalty factors $F_j = 1 - k_j s_j^2$, with severities $s_j \in [0, 1]$ and coefficients $k_1 = 0.12$ (trafficking and bonded child labour), $k_2 = 0.10$ (state-attributable civilian casualties), and $k_3 = 0.08$ (biospheric toxicity and PM2.5 exposure). The penalized index is $W^* = W \cdot \prod_{j=1}^{3} F_j$. The maximum mechanical deduction is $1 - 0.88 \times 0.90 \times 0.92 = 0.2706$, i.e. approximately 27.06%, which sits below the 30% governance cap on discretionary deductions; the cap is retained as a redundant invariant. The PM2.5 exposure threshold entering s_3 is set at **25 µg/m³**, exactly five times the 2021 WHO annual guideline of 5 µg/m³ (WHO, 2021; IQAir, 2021).

4.5 Sustainable Improvement Factor

Let $\Delta_c = (W^*_{c,t} - W^*_{c,t-3})/3$ be the annualized three-year change, $z_c = \text{clip}((\Delta_c - \mu)/\sigma, -2, 2)$, and $I_c = 1 + z_c/4 \in [0.5, 1.5]$. With the level ratio $L_c = W^*_{c,t}/\bar{W}^*_{world}$, the Sustainable Improvement Factor is

$$\text{SIF}_c = 0.70\, L_c + 0.30\, I_c.$$

SIF is published alongside W^* and I_c; it is undefined for countries lacking a comparable three-year series and is not bounded above.

4.6 Wellbeing Over Income (WOI)

A robust Huber M-estimator is fit to $\ln(W^*_c) = \beta_0 + \beta_1 \ln(\text{GDPpc_PPP}_c) + e_c$. The index is $\text{WOI}_c = 50 + 10\,(e_c - \text{median}(e))/\text{MAD}(e)$. WOI is descriptive, not causal; it may be negative or exceed 100; it is undefined when $\text{MAD}(e) = 0$; and full regression diagnostics are published with every release. Its purpose is to distinguish societies that convert income into wellbeing efficiently from those that do not, while the robust regression limits the leverage of outliers.

5 Psychometric and Robustness Standards

Because pillar 1 includes survey-based family and safety items, the validation programme imposes classical psychometric standards. Confirmatory factor analysis is conducted separately within six pre-specified cultural-linguistic clusters, with a minimum sample of 300 respondents per country and 1,000 for differential item functioning (DIF) testing. Measurement invariance must hold for at least 50% of survey indicators before cross-country comparisons are published (proposition P3 below). Sensitivity of rankings to goalpost choices is tested by comparing the 2nd/98th percentile specification against the 5th/95th percentile specification (proposition P5), and all robustness outputs are released with the data pipeline.

6 Capture-Resistant Governance

The governance design assumes that any measurement institution can be captured, and builds against that assumption from the outset.

Bootstrap stewardship. A single caretaker steward operates for no more than 24 months, with explicitly enumerated non-powers, public RFC processes, and an immutable methodological core during the bootstrap period, followed by a mandatory sunset of the founder's role.

Methodology Council. A seven-member council with staggered terms and a limit of two consecutive terms controls methodological changes.

Citizens' Assembly. An Assembly of approximately 500 members selected by stratified sortition holds an absolute, non-waivable veto over pillar definitions and weights — the normative heart of the index is thus owned by randomly selected citizens, not by technocrats, funders, or states.

Funding firewall. No single funder may contribute more than 20% of total funding in any cycle; violations freeze disbursement pending audit.

Data integrity. A three-member Emergency Data Integrity Team (EDIT) may impose a 72-hour publication hold on suspect data releases; appeals must be resolved within a 60-day window by an independent reviewer and, if unresolved, by a tribunal. Release integrity is anchored via OpenTimestamps, which establishes existence and ordering of hashes — not truth, authorship, or legitimacy, which remain matters for review.

Forkability. The specification text is licensed CC BY 4.0 and reference code Apache 2.0, so that if the central institution ever deviates from its charter, the scholarly community can fork the entire framework and continue it independently.

7 Validation Programme and Falsifiability

The framework makes falsifiable commitments and defines its own failure conditions. The validation propositions are: (P1) auditable coverage of at least 120 countries; (P2) dual-source independence for data feeds; (P3) measurement invariance for at least 50% of survey indicators; (P4) successful replication of the full pipeline by two independent teams with a third-party adjudicator; and (P5) robustness of rankings across goalpost sensitivity tests. Broader triggers include persistent rank instability, demonstration that the index is effectively a relabeling of GDP, mathematical infeasibility of the penalty regime, or governance capture.

A 36-month kill-switch halts or forces redesign of the programme if coverage, source independence, or invariance thresholds are not met, with two independent audits and a council supermajority required to proceed. The pilot schedule runs from 2026 (preregistration and pilot build) through 2027 (two-team replication with a third adjudicator) to 2028 (release, revise, or no-go decision). A published failure report is itself treated as a publishable scientific outcome.

Financial firewall. The WCI involves no legal-tender designation, no peg, no redemption promise, no amendment of the SDR, and no automatic sanctions; WCI-linked financial instruments carry an explicit force-majeure clause (WCI Master Specification, §7). The index is a measurement and research construct, and nothing in this specification should be read as monetary-policy advocacy.

8 Invitation to Global Collaboration

This specification is complete enough to be implemented, criticized, and falsified — and incomplete enough to need the world's help (WCI Master Specification, §1). The author therefore issues an open, standing invitation to mathematicians and statisticians, monetary and financial economists, development and family sociologists, peace and conflict scholars, jurists, psychologists and psychometricians, philosophers, data scientists, and citizens' assemblies worldwide: to scrutinize the mathematics, contest the weights, stress-test the governance

design, propose better indicators, replicate the pipeline on independent data, and — where conviction warrants it — fork the framework and build better successors.

No scientific consensus can or should be declared by a single author. The goal of this release is precisely to initiate that consensus process in the open: public commentary, adversarial replication, methodological debate, and convergent refinement toward a measurement of human flourishing worthy of the name. Contributions will be acknowledged through the open-versioning and attribution mechanisms of the repository hosting this work; the licensing (CC BY 4.0 / Apache 2.0) guarantees that every improvement remains permanently open to all.

9 Limitations and Conclusion

Three classes of limitation are acknowledged openly (Talebian Rizi, 2026a). First, any published index becomes a target: Goodhart's and Campbell's laws predict that once governments are ranked, measurement gaming begins (Goodhart, 1975). Multi-source auditing and the citizens' veto mitigate but cannot eliminate this. Second, the weights — especially the 50% weight on family and human development — are a normative, contestable commitment, defended here on sociological grounds but open to democratic revision through the Assembly; the goalpost sensitivity machinery exists precisely so that critics can quantify the consequences of disagreeing. Third, the geometric unit floor compresses the bottom tail, WOI is descriptive rather than causal and can mislead when income and infrastructure access diverge, and SIF is unbounded above. This document is a specification, not an empirically validated ranking; it makes no causal claims and offers itself for refutation. If the framework fails its own tests, the honest publication of that failure will itself have been worth the effort.

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